MATHEUS TEIXEIRA



Machine Learning | Data Science | Computer Vision | Software Development

CONTACT

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SKILLS

Programming

Python
C/C++
Java
JavaScript/TypeScript
Rust
HTML5/CSS
ASP
SQL
LaTeX

Machine Learning & Data

PyTorch
scikit-learn
TensorFlow/Keras
pandas/NumPy/SciPy
Matplotlib/Seaborn
Computer Vision
Model Evaluation

Web & Software

React
Node.js/Electron
Three.js/WebAssembly
Playwright/Vitest/pytest

Quantum Machine Learning

Qiskit
QSVM
Quantum Kernels
IBM Quantum

Tools & Systems

Git
Linux

Languages

Portuguese
English (C2)

PROJECTS

- Amadeo: sound to sheet music
- Isaac: 3D physics simulations
- Allwinner: crypto signal backtesting
- Brectus: AI coding agent orchestrator

PUBLICATIONS

- Systematic Review of Quantum Machine Learning for Healthcare – BRACIS 2026

PROFILE

Applied Machine Learning Researcher with an M.Sc. in Electrical Engineering, an enrolled CAPES-funded Ph.D. student in Electrical Engineering, and currently pursuing an MBA in Data Science, Artificial Intelligence and Analytics. Experienced in reproducible machine learning, computer vision, model evaluation, healthcare AI, and quantum machine learning, with doctoral research on quantum error correction. Also builds tested full-stack applications in TypeScript, React, Rust/WebAssembly, Python and Node.js.

EXPERIENCE

Applied Machine Learning Researcher

03/2024 – 03/2026 FEI, São Bernardo do Campo - SP

Developed a reproducible pipeline for breast cancer histopathology image classification using BreaKHis, ResNet-50, patient-wise splitting, and SVM/QSVM models, achieving 80.0% accuracy, 86.0% F1-score, and an MCC of 0.520 with the QSVM pipeline.

Conducted experiments on IBM Quantum hardware and analyzed the effect of noise on quantum kernels under NISQ conditions.

Web Developer and IT Support Assistant

04/2016 – 07/2023 Orion Benefícios, Santos - SP

Developed and maintained the company's website using HTML, CSS, ASP, and database-integrated systems, implementing improvements and maintaining legacy code.

Provided technical support for computers, software, printers, networks, infrastructure, and database systems.

Teaching Assistant in Analytical Geometry

08/2020 – 12/2020 FEI, São Bernardo do Campo - SP

Helped students with matrices, linear systems, vectors, and lines and planes through help sessions and exercise lists, in the semester after taking the course.

EDUCATION

Ph.D. in Electrical Engineering

02/2027 – 02/2030 FEI, São Bernardo do Campo - SP

CAPES Research Scholarship Recipient. Enrolled; classes begin in February 2027. Focus on quantum error correction, machine learning decoders and fault-tolerant quantum computing; sandwich year planned at ETH Zurich (2028–2029).

MBA in Data Science, Artificial Intelligence and Analytics

08/2026 – 08/2027 USP, Piracicaba - SP

Full Scholarship Recipient. Currently pursuing the program with a focus on statistics, Python, data analysis, machine learning, artificial intelligence, and analytics.

M.Sc. in Electrical Engineering

03/2024 – 03/2026 FEI, São Bernardo do Campo - SP

CAPES Research Scholarship Recipient. Concentration in Artificial Intelligence Applied to Automation and Robotics. Dissertation on quantum machine learning applied to breast cancer image analysis.

B.Sc. in Computer Science

07/2019 – 06/2023 FEI, São Bernardo do Campo - SP

Full Scholarship Recipient. Undergraduate thesis on machine learning and quantum computing using a quantum classifier applied to the Iris dataset.

Technical Diploma in Building Construction

02/2016 – 06/2017 ETEC Aristóteles Ferreira, Santos - SP

Technical training in building construction, combining theoretical knowledge with practical project development.